



**BETTER
SOFTWARE
BETTER
RESEARCH**

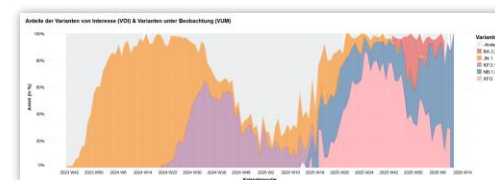
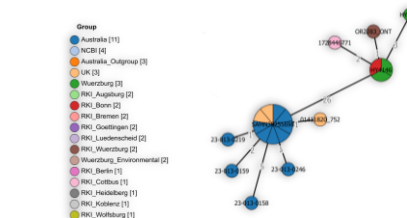
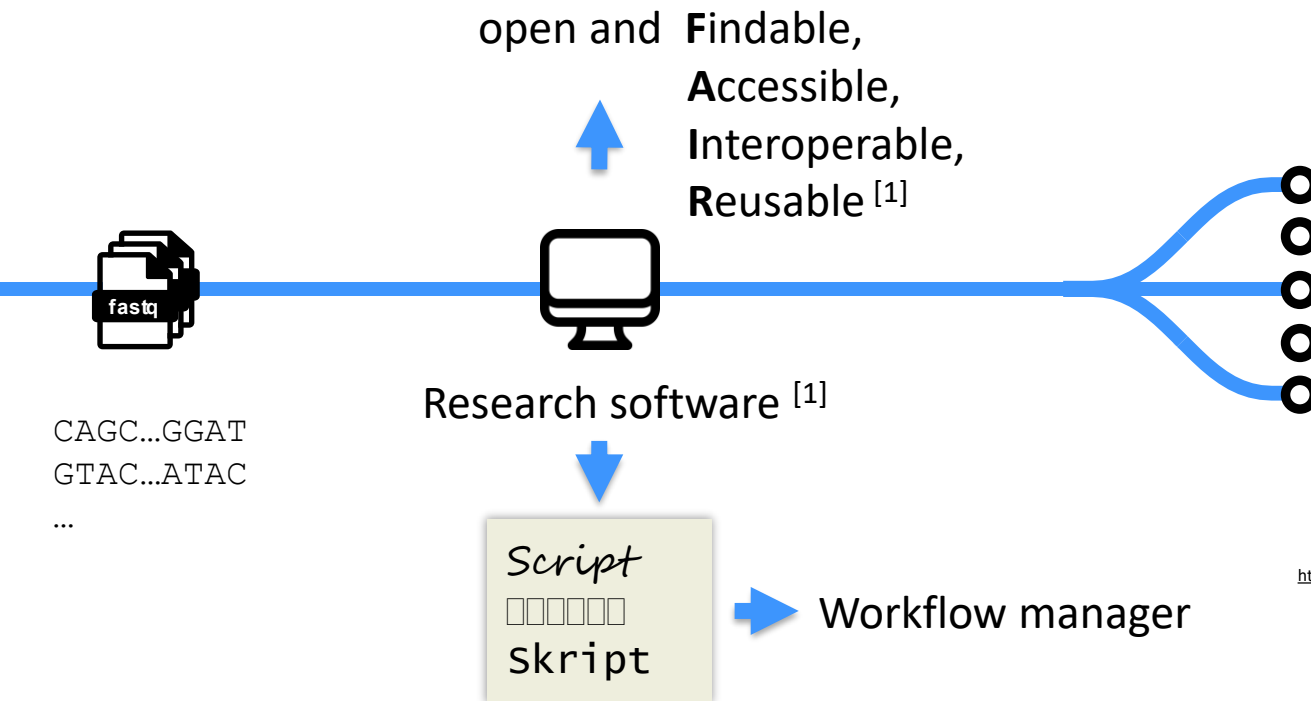
Interlude: From scripts to workflows

**A short introduction into Research
Software Engineering (RSE)**

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Research Software Engineering



https://public.data.rki.de/t/public/views/IGS_Dashboard/DashboardVOC, 2026-04-09

[1] Barker *et al.* 2022. 10.1038/s41597-022-01710-x



Workflow Managers

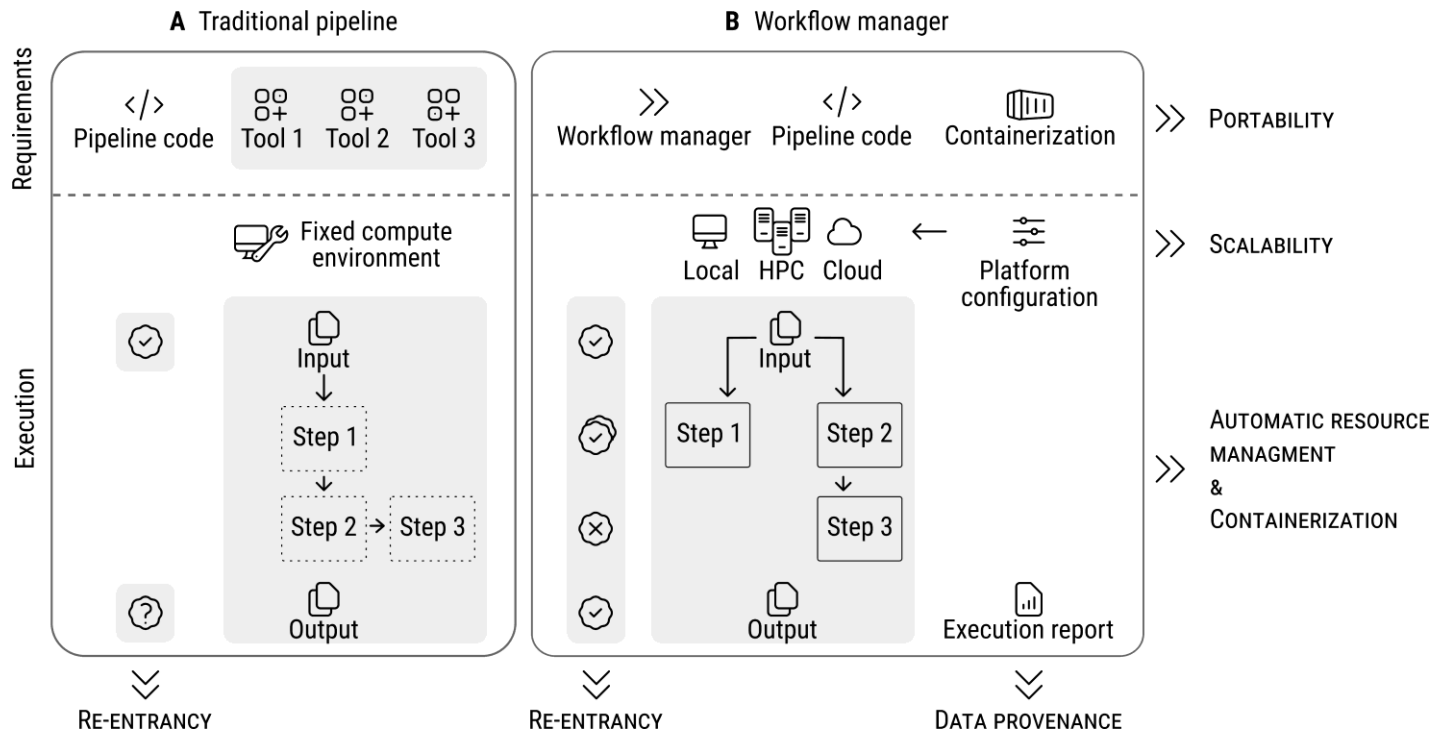
“Workflow managers provide a framework for the creation, execution, and monitoring of pipeline. [...]

They simplify pipeline development, optimize resource usage, handle software installation and versions, and run on different compute platforms, enabling workflow portability and sharing.”^[2]

^[2] Wratten, L., Wilm, A. & Göke, J. Reproducible, scalable, and shareable analysis pipelines with bioinformatics workflow managers. *Nat Methods*. 2021.
<https://doi.org/10.1038/s41592-021-01254-9>



Traditional Pipelines vs. Workflow Manager



Adapted from Wratten *et al.* 2021. 10.1038/s41592-021-01254-9

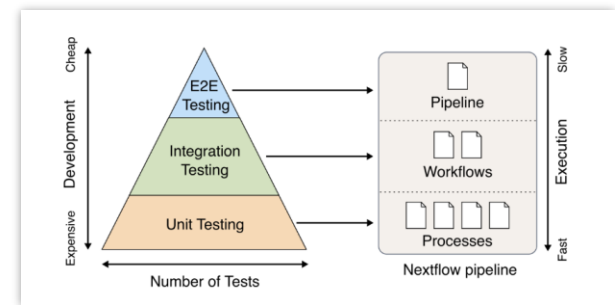


Workflow Manager: nextflow

- ✓ Automatic resource management
- ✓ Re-entrancy
- ✓ Data- and pipeline provenance
- ✓ Scalability, portability, containerization
- ✓ Modularization
- ✓ nf-core community
 - ✓ Tools, templates, linting, trainings
 - ✓ Curated and standardized resources :
 - ✓ > 132 pipelines
 - ✓ > 78 subworkflows
 - ✓ > 1521 modules
- ✓ Test-framework: nf-test



nf-core 



Forer et al. 2024. 10.1101/2024.05.25.595877